STUDY GROUP ON FUTURE DATA INFRASTRUCTURE (SG-FIT)

Move Less – Use More: tech demo

DEMO SCRIPT https://github.com/wmo-im/metdx-demo/blob/main/docs/demo/infcom4-sg-fit-side-event-2026-11-25.md

WORLD METEOROLOGICAL ORGANIZATION

Step 1 — Find the dataset

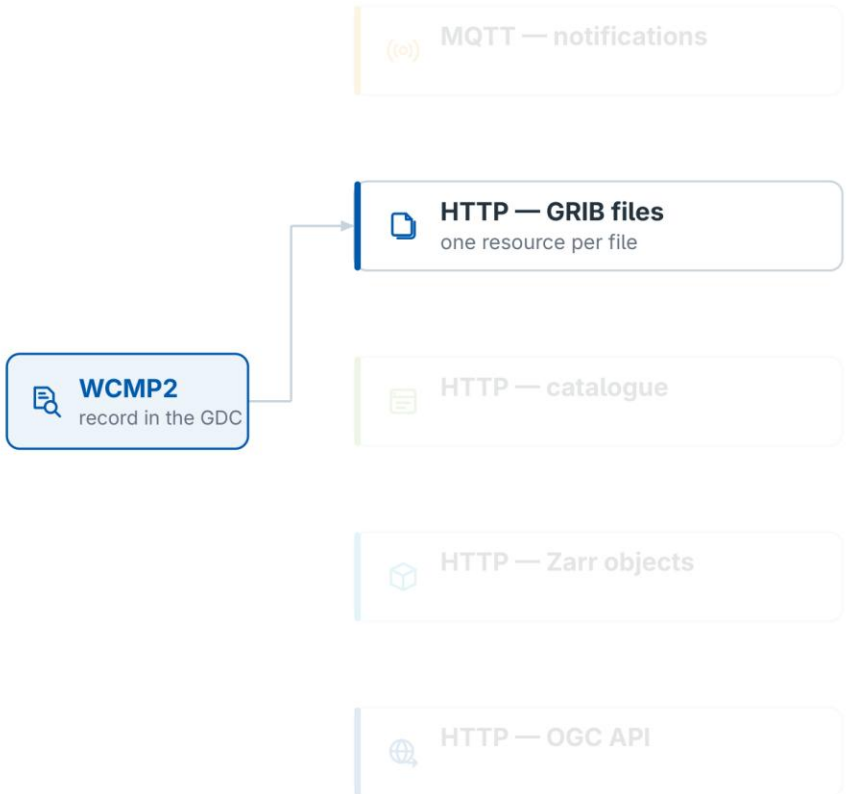
Discovery metadata is the entry point to everything that follows



- A **WCMP2** discovery metadata record is published to the WIS2 **Global Discovery Catalogue** — an OGC API – Records service.
- It gives a user enough to *find* the dataset and decide whether it meets their requirements.
- It also says how to get the data: download, API, or subscribe.
- If it is not in the catalogue, users cannot find it.

Step 2 — The data is a pile of GRIB files

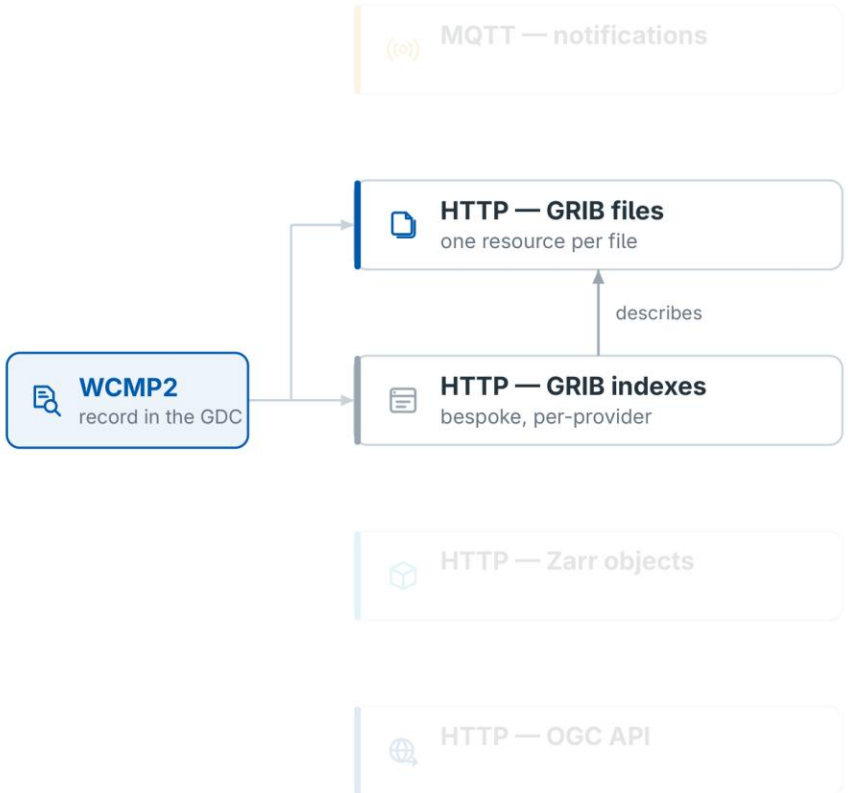
This is, after all, WMO



- WIS2 is Web-oriented, so the GRIB files are published as **HTTP resources**. The WCMP2 record may point at a Web Accessible Folder.
- WIS2 does *not* formalise how resources are grouped or organised, and does not require a file-naming convention.
- So the user has to navigate whatever the data provider chose.
- And a single dataset can be thousands of individual resources to manage: a significant overhead.

Step 3 — Which file do I actually need?

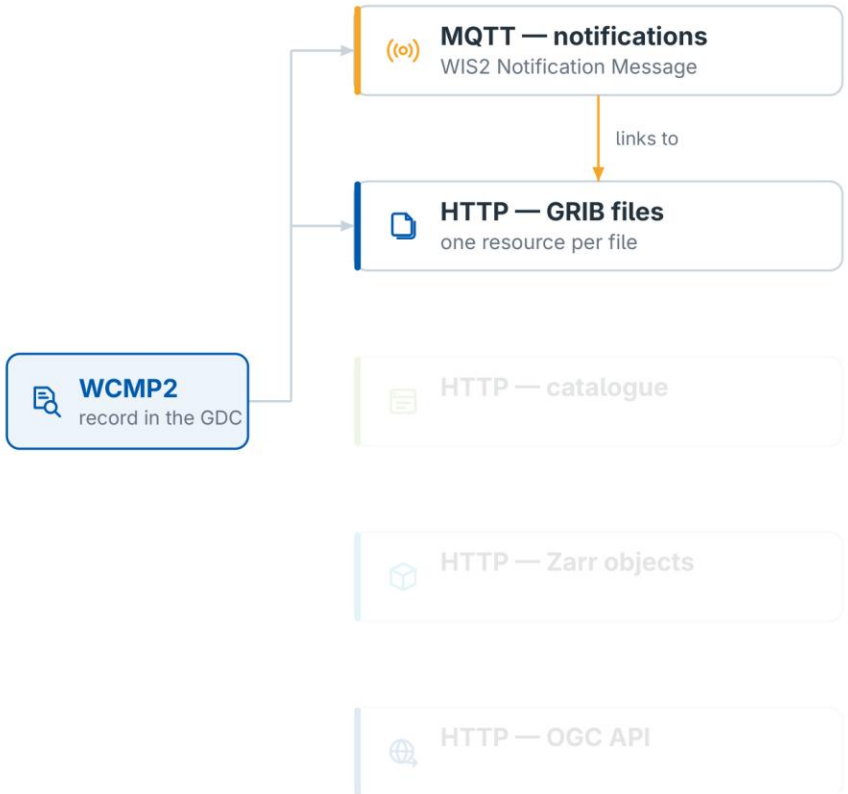
The metadata is inside the GRIB, which is the wrong place for it



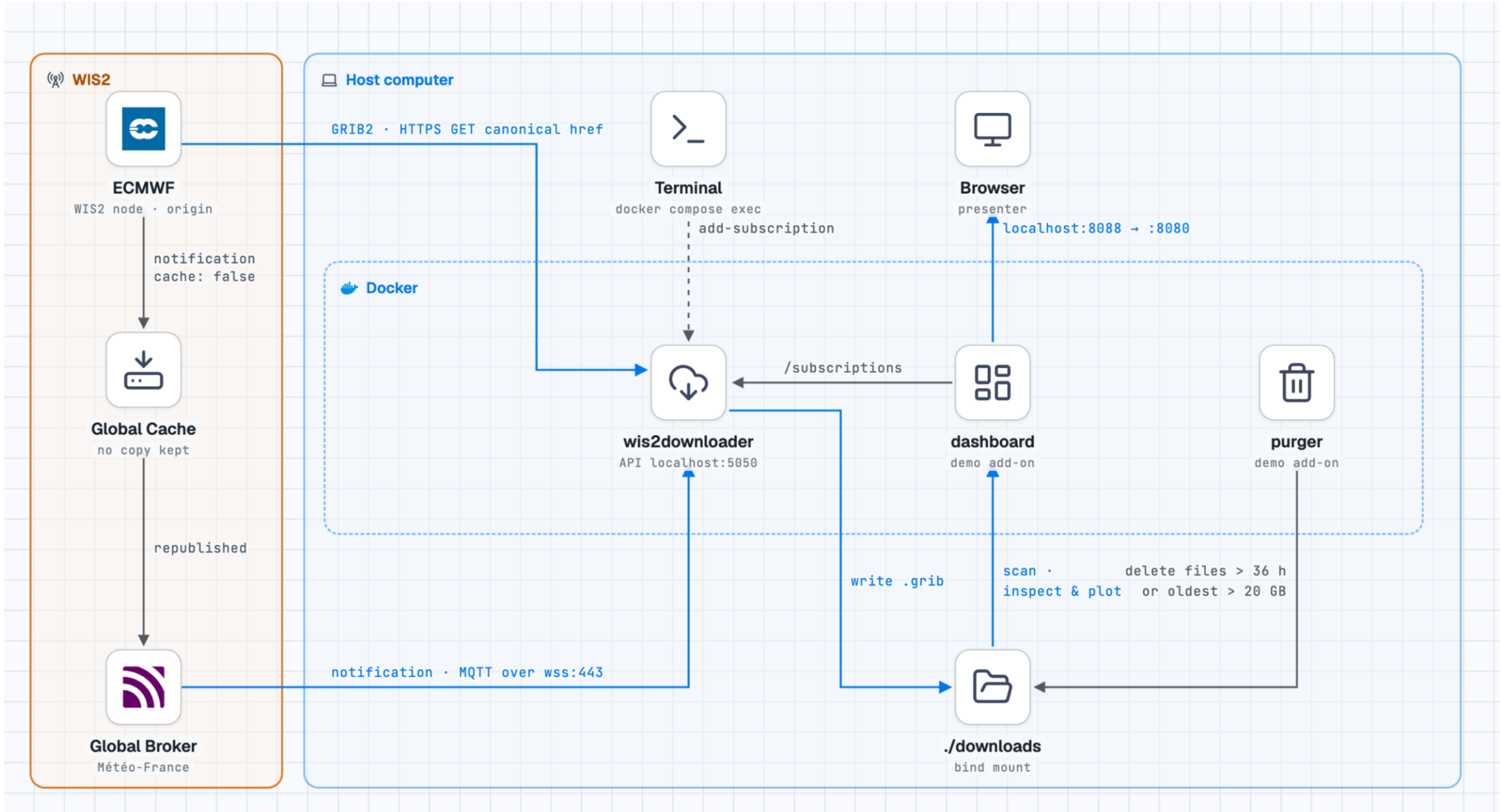
- To know what a file contains you must open and parse it. Fine on local POSIX storage; painful for objects on cloud storage.
- **ECMWF** and **NOAA** both recognised this and independently adopted data models, tooling and services that publish *indexes* describing each GRIB.
- It works — but it is not standardised. Different tooling, different formats, different semantics.

Step 4 — Real-time: publish / subscribe

WIS2 replaces push-by-routing-table with subscribe-to-what-you-need

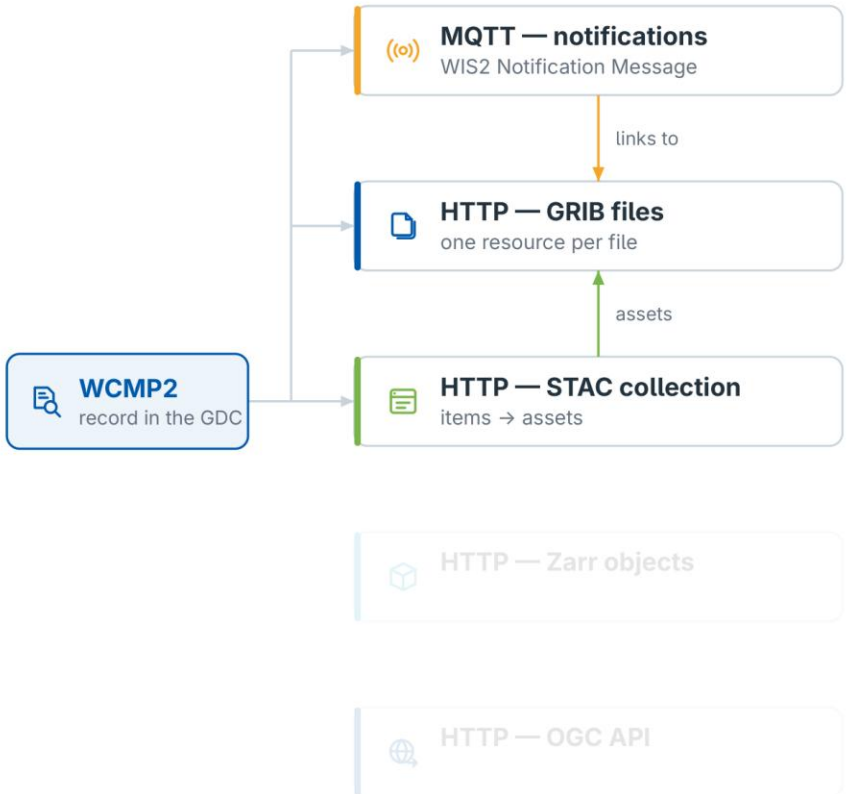


- WIS2 adds **Publish-Subscribe (Pub/Sub)** over MQTT, with a standard Topic Hierarchy and a GeoJSON Notification Message.
- Each notification message describes a newly available resource — date, time, location, topic — and links to the data itself.
- The WIS2 Cookbook has a recipe for richer metadata to support **client-side filtering**, and for declaring those filters in WCMP2.
- Users subscribe to the topics they care about and download only what they need.

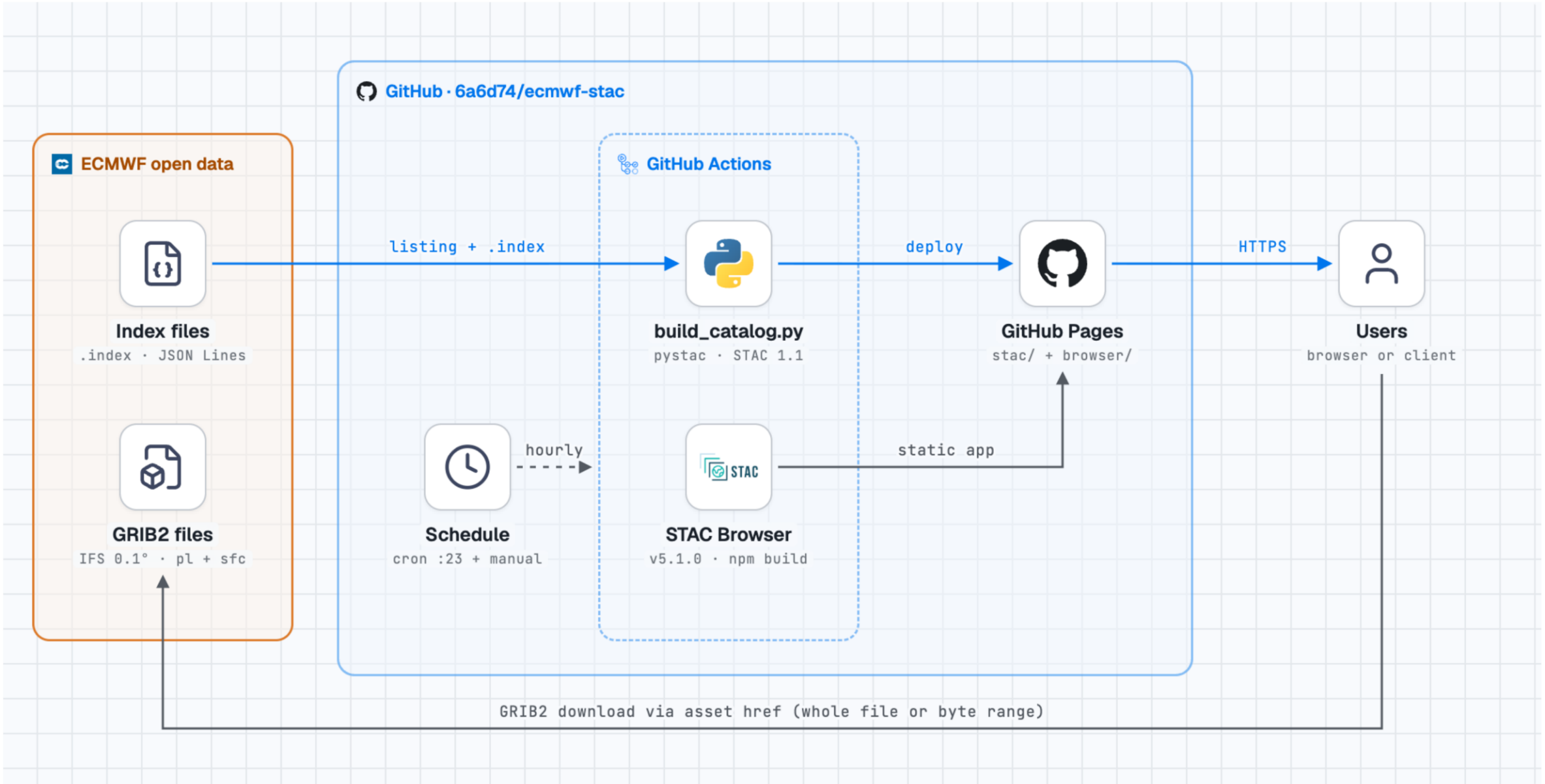

WIS2
Host computer
ECMWF
WIS2 node · origin
GRIB2 · HTTPS GET canonical href
Terminal
docker compose exec
add-subscription
Browser
presenter
localhost:8088 → :8080
Docker
notification
cache: false
Global Cache
no copy kept
/subscriptions
wis2downloader
API localhost:5050
dashboard
demo add-on
purger
demo add-on
republished
write .grib
scan ·
inspect & plot
delete files > 36 h
or oldest > 20 GB
notification · MQTT over wss:443
Global Broker
Météo-France
./downloads
bind mount

Step 5 — Standardise the index: STAC

Make a pile of files easy to navigate

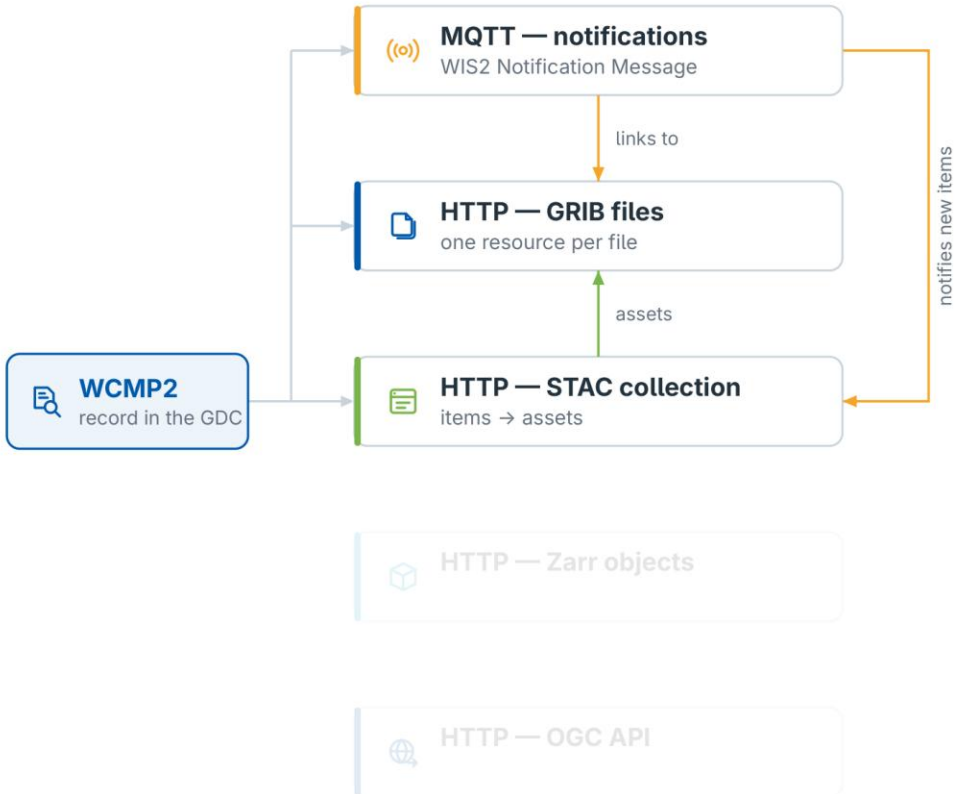


- STAC — the SpatioTemporal Asset Catalog, an OGC community standard — is a common way to browse collections of geospatial data resources, whatever format the data is in.
- Collections organise data into virtual folders; items carry resource-level metadata and point at **assets** — the data resources themselves.
- Wide adoption and good open-source tooling. Providers could publish STAC alongside their resources.
- We've built one for ECMWF's open data, generated hourly from ECMWF's index files, without downloading a GRIB.


GitHub · 6a6d74/ecmwf-stac
ECMWF open data
GitHub Actions
Index files
.index · JSON Lines
listing + .index
build_catalog.py
pystac · STAC 1.1
deploy
GitHub Pages
stac/ + browser/
HTTPS
Users
browser or client
GRIB2 files
IFS 0.1° · pl + sfc
Schedule
cron :23 + manual
hourly
STAC
STAC Browser
v5.1.0 · npm build
static app
GRIB2 download via asset href (whole file or byte range)

Step 6 — One message per file is a lot of messages

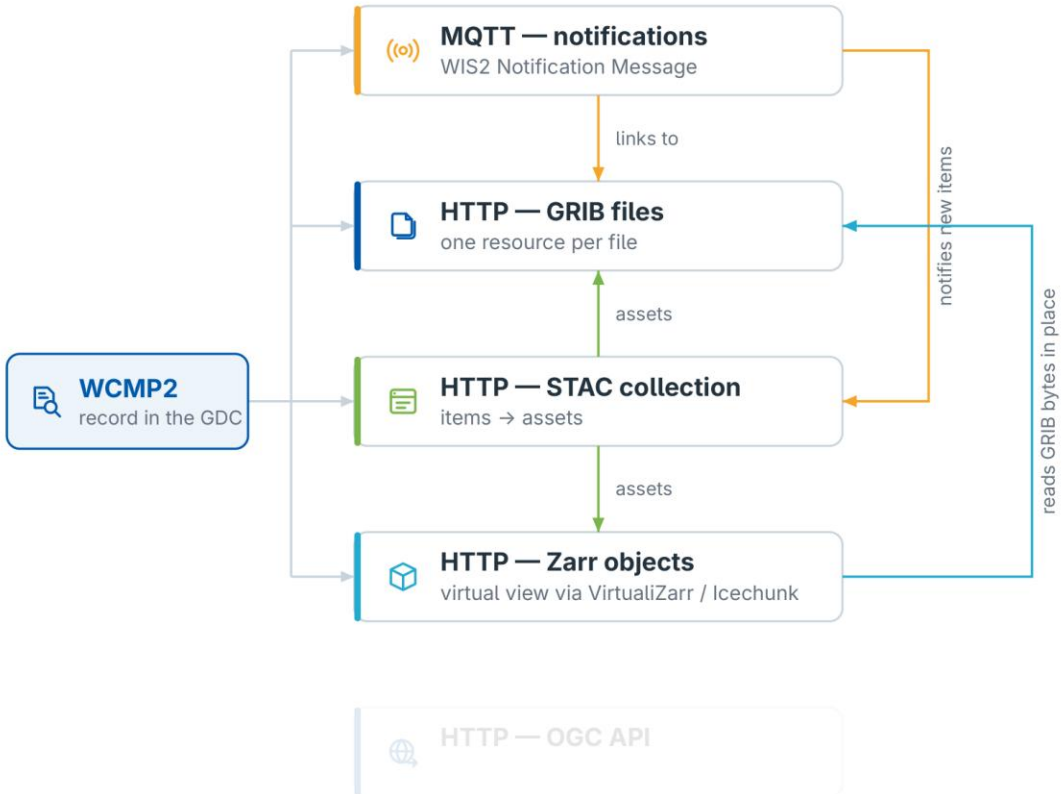
A notification per model run, not per file



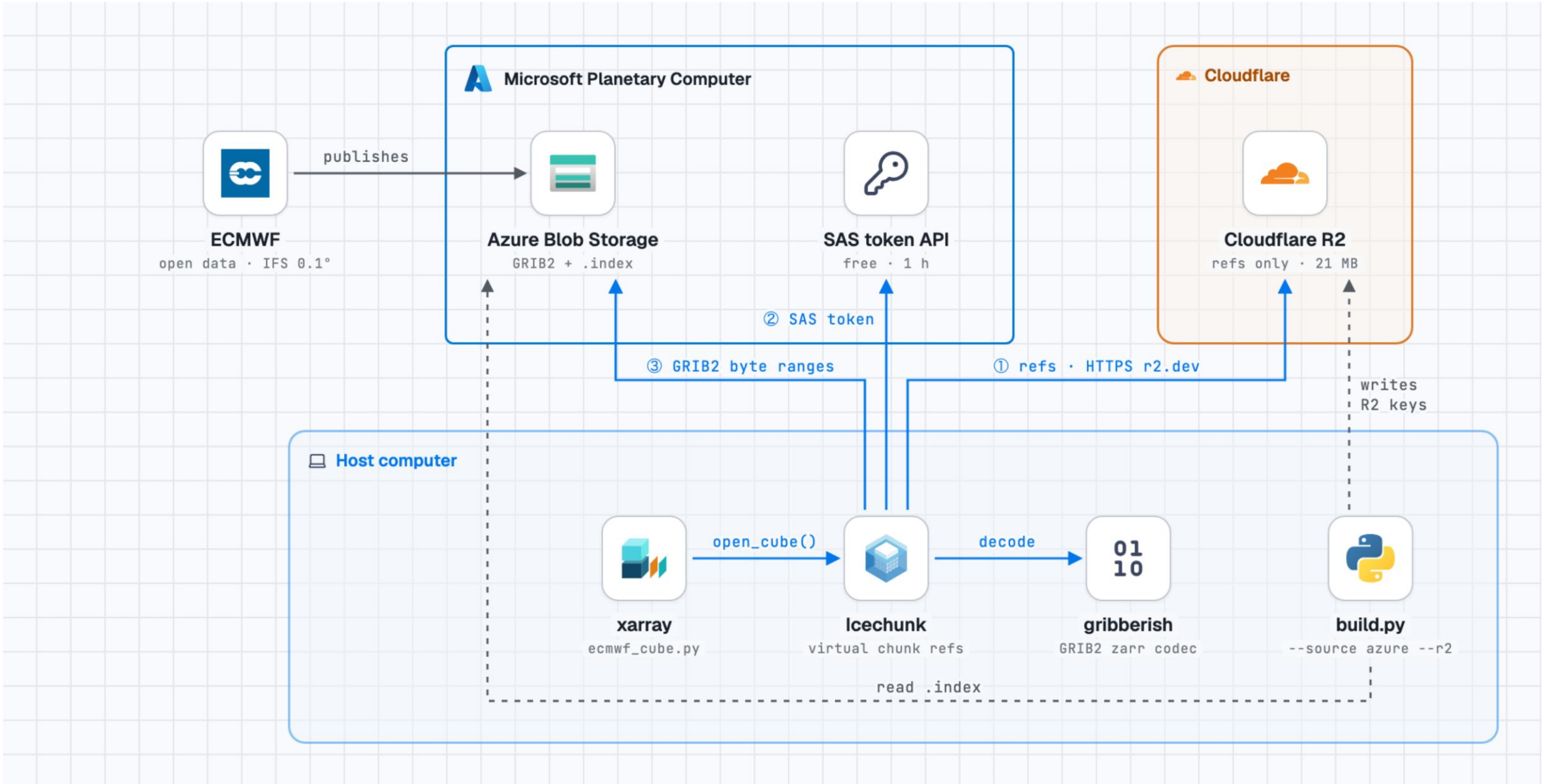
- One WIS2 Notification Message per GRIB file. For a large model run, that is a great many messages for a user to receive and filter.
- With a standard index, one message could announce a new STAC item, such as a whole model run, and the user follows it to the GRIBs they need.
- For early access, republish the item with "rel": "update" as more files arrive.

Step 7 — Work at the dataset level: virtual Zarr

Let the software figure out which bytes a calculation needs

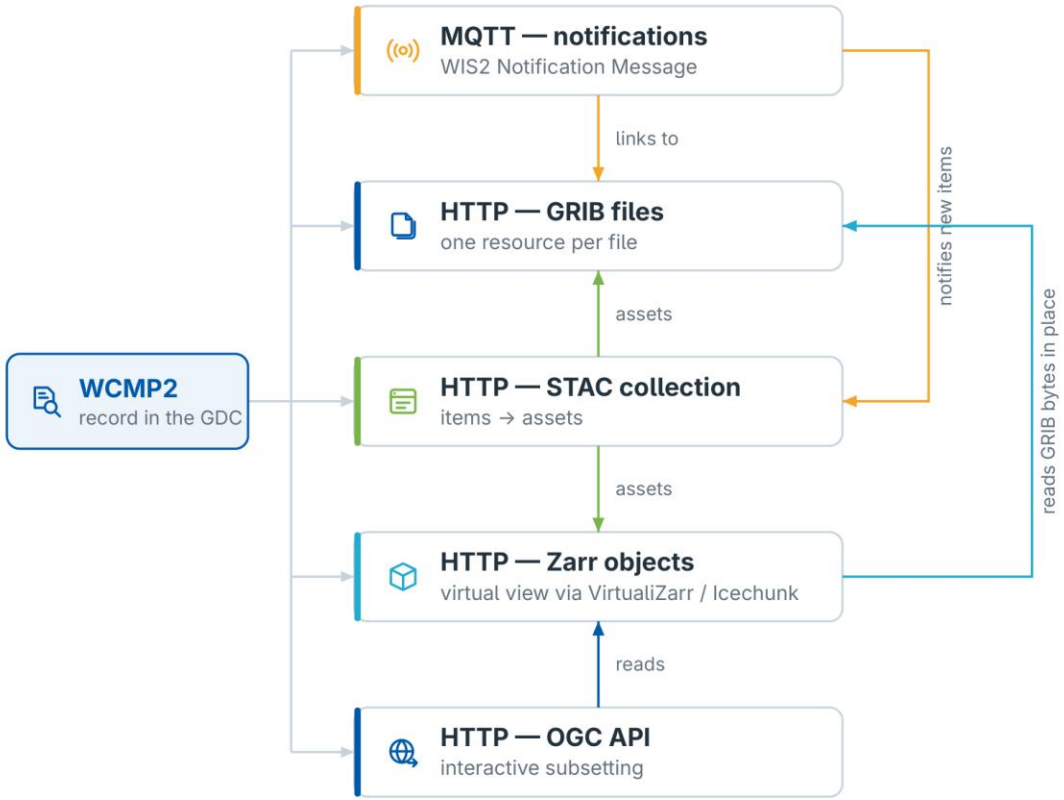


- Users would rather not handle thousands of resources. Cloud-optimised formats such as Zarr let you open a whole dataset lazily: `ds = xarray.open_zarr(…)` — bytes are fetched only when needed.
- Native Zarr is one option. Given WMO's continuing need for legacy formats, VirtualiZarr / Icechunk are attractive: the data stays in its native format, the user sees a Zarr store, and an index says which bytes to read.
- STAC collections can carry Zarr assets alongside GRIB assets, so users find the store they need.
- We've built one: an Icechunk cube of ECMWF's 0.1° forecasts; the GRIBs stay where ECMWF publishes them.

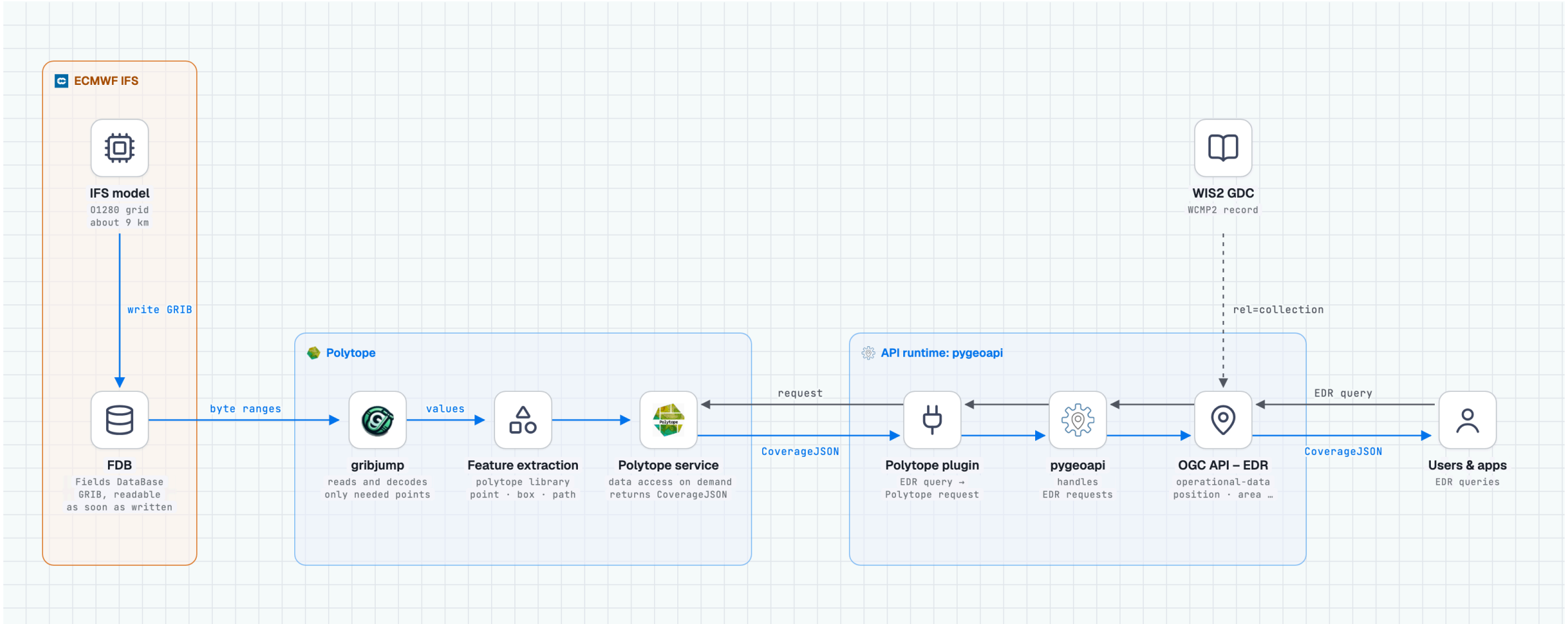

Microsoft Planetary Computer
Cloudflare
publishes
ECMWF
open data · IFS 0.1°
Azure Blob Storage
GRIB2 + .index
SAS token API
free · 1 h
Cloudflare R2
refs only · 21 MB
② SAS token
③ GRIB2 byte ranges
① refs · HTTPS r2.dev
writes
R2 keys
Host computer
open_cube()
decode
01
10
xarray
ecmwf_cube.py
Icechunk
virtual chunk refs
gribberish
GRIB2 zarr codec
build.py
--source azure --r2
read .index

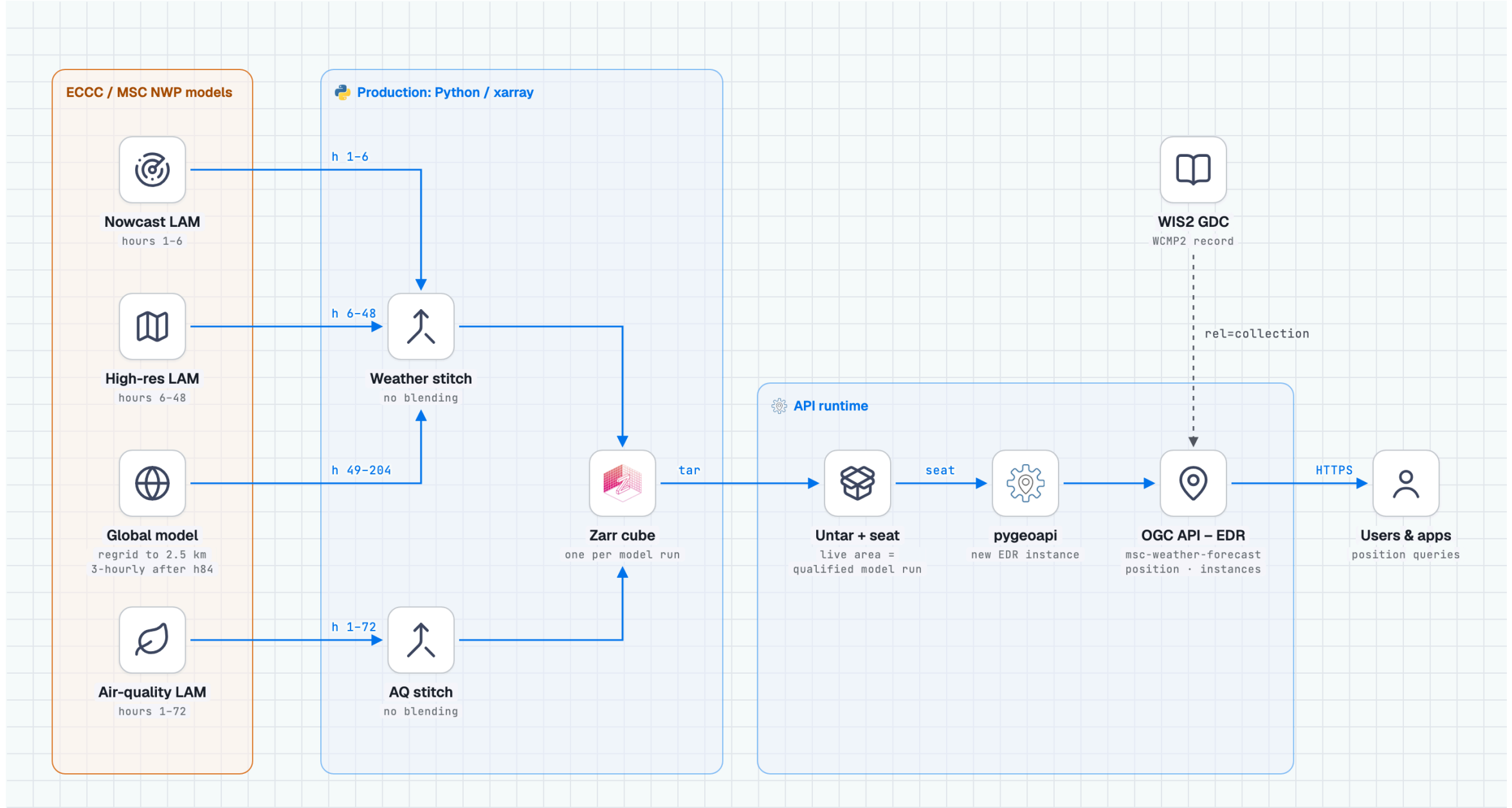
Step 8 — For the rest of us, a Web API

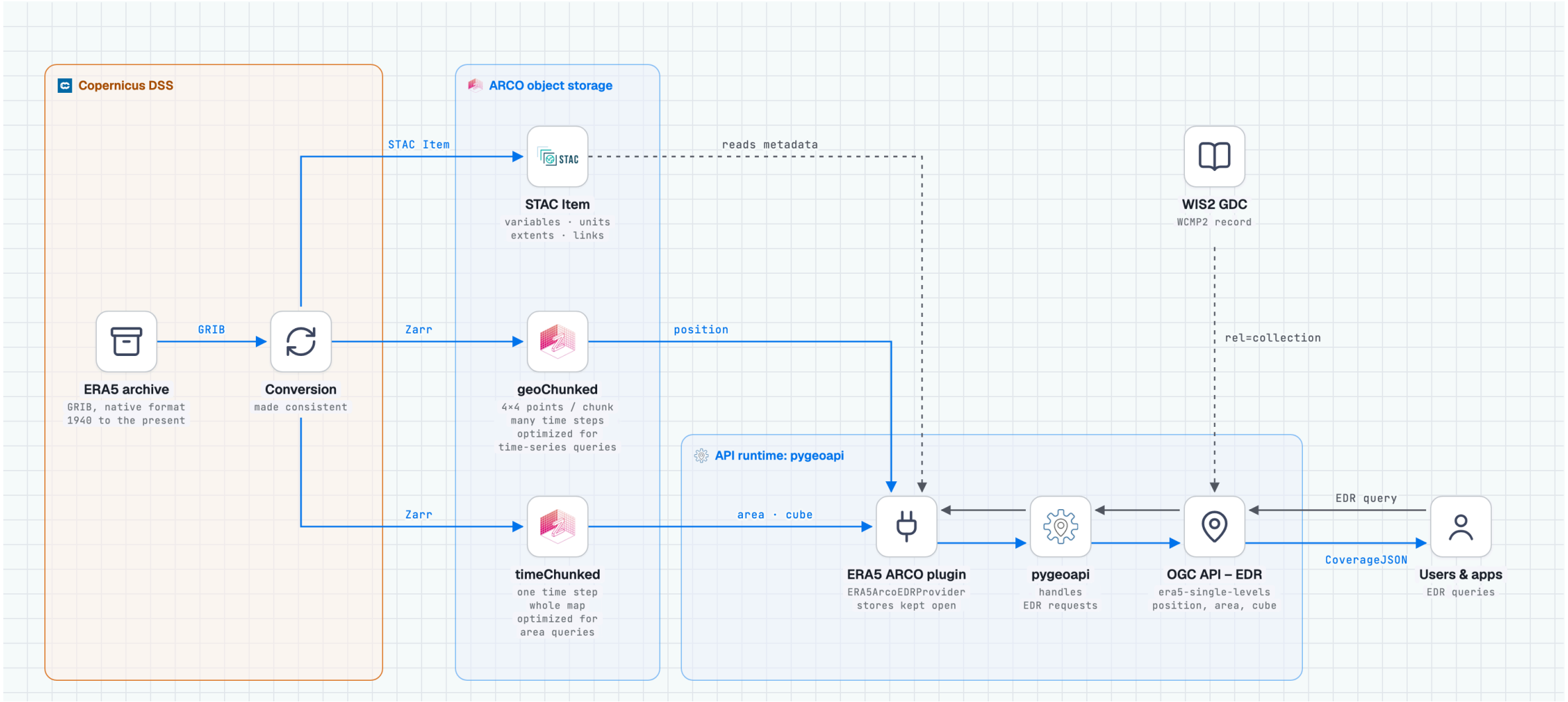
Most users want a small subset/extract or graph/image



- Zarr suits power users willing to learn how a dataset is structured. Many users just want a snippet for a Web application.
- An HTTP **GET** meets that need: the subset, in the format they asked for.
- EUMETNET MeteoGate already demonstrates this — interactive Web services on WIS2 foundations.
- How the request is served is the provider's choice, and there is always a cost — especially at scale. Cloud-optimised data is a good foundation. There are many options to implement an OGC API: pygeoapi (Xarray by default); pygeoapi with custom tooling (ECMWF Polytope and Earthkit); IBL; ...
- We've built a Web UI to demonstrate how user applications can interact with Web APIs in a metadata-driven workflow.

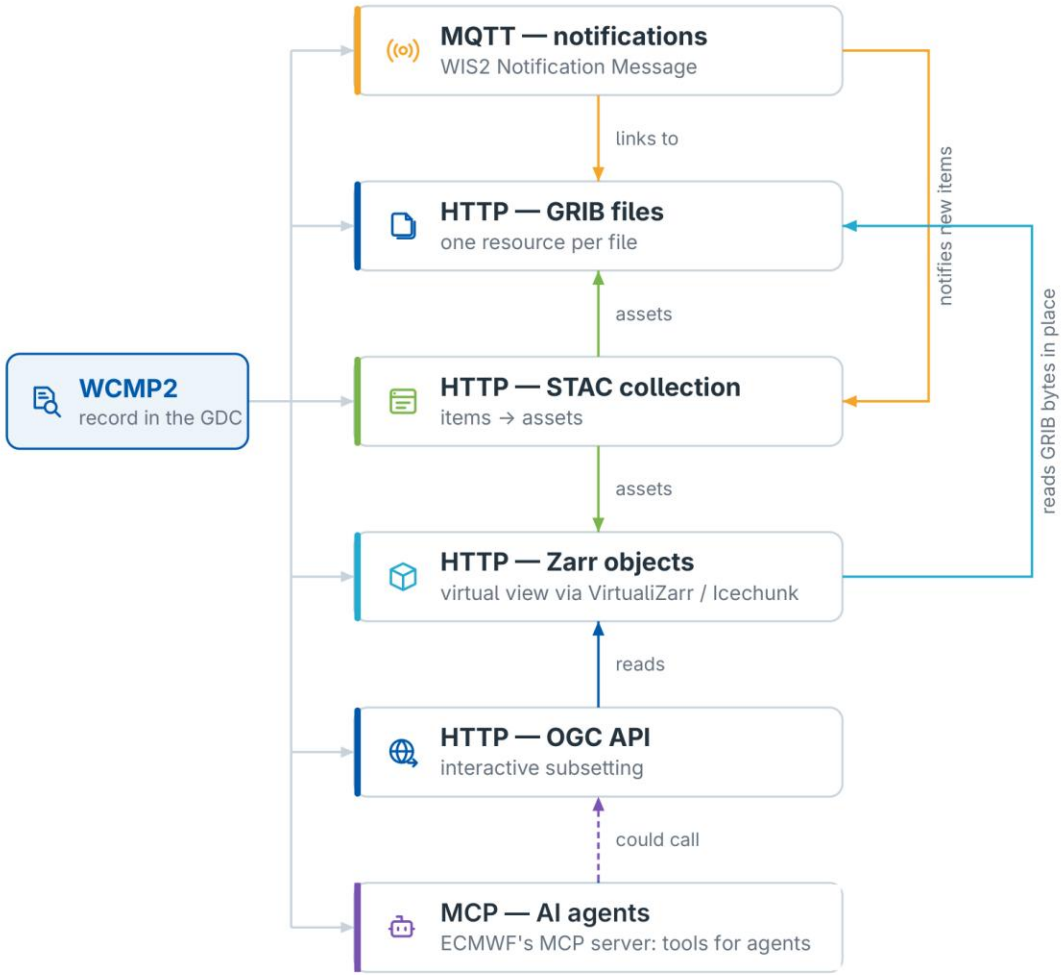

ECMWF IFS
IFS model
O1280 grid
about 9 km
write GRIB
FDB
Fields DataBase
GRIB, readable
as soon as written
byte ranges
Polytope
gribjump
reads and decodes
only needed points
values
Feature extraction
polytope library
point · box · path
Polytope service
data access on demand
returns CoverageJSON
request
CoverageJSON
API runtime: pygeoapi
Polytope plugin
EDR query →
Polytope request
pygeoapi
handles
EDR requests
OGC API – EDR
operational-data
position · area …
WIS2 GDC
WCMP2 record
rel=collection
EDR query
CoverageJSON
Users & apps
EDR queries


ECCC / MSC NWP models
Nowcast LAM
hours 1-6
High-res LAM
hours 6-48
Global model
regrid to 2.5 km
3-hourly after h84
Air-quality LAM
hours 1-72
Production: Python / xarray
h 1-6
h 6-48
h 49-204
h 1-72
Weather stitch
no blending
AQ stitch
no blending
Zarr cube
one per model run
tar
API runtime
Untar + seat
live area =
qualified model run
seat
pygeoapi
new EDR instance
WIS2 GDC
WCMP2 record
rel=collection
OGC API – EDR
msc-weather-forecast
position · instances
HTTPS
Users & apps
position queries


Copernicus DSS
ARCO object storage
STAC Item
reads metadata
WIS2 GDC
WCMP2 record
GRIB
Zarr
position
rel=collection
ERA5 archive
GRIB, native format
1940 to the present
Conversion
made consistent
STAC Item
variables · units
extents · links
geoChunked
4×4 points / chunk
many time steps
optimized for
time-series queries
API runtime: pygeoapi
Zarr
area · cube
EDR query
CoverageJSON
timeChunked
one time step
whole map
optimized for
area queries
ERA5 ARCO plugin
ERA5ArcoEDRProvider
stores kept open
pygeoapi
handles
EDR requests
OGC API – EDR
era5-single-levels
position, area, cube
Users & apps
EDR queries

And beyond — integration with AI agents

Let an AI agent follow the links for you



- AI agents can follow the same metadata-driven workflow. The WCMP2 record links to ECMWF's MCP server: the Model Context Protocol, an open standard for agents to discover and use a provider's tools.
- An agent such as Claude finds the link in the record, connects, and answers in plain language: "Show me the ECMWF meteogram for Geneva."
- Today the server returns ECMWF's charts, maps and catalogue information. Next: agents requesting data through OGC APIs such as EDR.